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Do polycentric urban regions promote functional spillovers and economic performance? Evidence from China

Yixiao Wang^a, Bindong Sun^b  and Tinglin Zhang^c 

ABSTRACT

Although the literature often suggests that the economic prowess of 'polycentric urban regions' partly results from the ability of their constituent cities to promote functional spillovers through 'borrowed size', the empirical evidence remains scarce. This paper finds that polycentricity boosts the positive functional spillovers of producer service sector of city clusters beyond their boundaries only when city clusters have larger population and better urban infrastructure connectivity. And consequently, functional spillovers increase urban labour productivity. The labour division among cities could be the essential mechanism for the linkage between polycentricity and regional functional spillovers.

KEYWORDS

polycentric urban regions; functional spillovers; labour productivity; borrowed size; labour division; producer service sector

JEL O18, R12

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INTRODUCTION

Aiming to promote coherence and competitiveness, polycentric spatial planning projects have been launched in Europe, such as the European Spatial Development Perspective (ESDP), POLYNET: Sustainable Management of European Polycentric Mega-City Regions (POLYNET) and European Spatial Planning Observation Network (ESPON). Partly influenced by European practice and policies, the polycentric spatial planning concept was implemented in China at the end of the 20th century to promote more sustainable and balanced development (Cheng & Shaw, 2018).

Accordingly, an increasing number of academic discussions are being conducted on polycentric urban regions (PURs). A PUR is defined as a set of urban centres without a dominant centre in the population (Parr, 2004). One of the important issues is their performance, especially economic performance, which is related to the economic rationality of the polycentricity policy. While some studies

pose the critique that polycentricity does harm to the economy of the agglomeration because of relatively poor geographical proximity, the limit of critical mass and the inconvenience in the circulation of flow (Ciccone, 2002; Lambooy, 1998; Parr, 2004). Other literature suggests that the economic potential of PURs results not only from the alleviation of the agglomeration diseconomies of monocentric metropolitan regions (Fujita et al., 1997; Parr, 2004) but also from positive network externalities through the 'borrowed size' effect among the constituent cities inside a PUR (Camagni & Salone, 1993; Capello, 2000; Meijers & Burger, 2010). Recently, some research finds the positive relationship between economic performance and polycentricity of Chinese city clusters (Sun et al., 2019; Wang et al., 2019).

The notion of 'borrowed size' can be traced back to Alonso (1973); it refers to the phenomenon by which a small city benefits from using the high-ranking infrastructure of a nearby larger city. This concept was later enriched and extended to the functional interactions between firms

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and populations in different cities (Meijers & Burger, 2017; Phelps et al., 2001). These spatial interactions were thought to form a functional city network, and the driving force was implicitly related to functional synergy, functional complementarity or the labour division among cities (Camagni & Salone, 1993; Van Oort et al., 2010). In contrast to the borrowed size effect, the agglomeration shadow effect has been proposed as the negative outcome of the interaction of network externalities in a single city. For example, the functions of some cities are greater than expected given their population size, and they enjoy the borrowed size effect; however, other cities have fewer functions relative to their size and suffer the agglomeration shadow effect within a region (Meijers & Burger, 2017). There is a debate about whether the borrowed size effect or the agglomeration shadow effect dominates inside PURs (Meijers & Burger, 2017). Only when the former surpasses the latter can PURs acquire positive regional functional spillovers or benefit from the positive influence of city network externalities and thus obtain economic benefits.

Despite the rich theoretical literature, there is little empirical evidence on the economic advantages of PURs through regional functional spillovers (Meijers & Burger, 2010). More evidence on this topic would hold great significance for supporting the economic rationale of PURs. To improve these empirical deficiencies, using 2004–14 panel data we examine the impact of the polycentricity of 13 city clusters in China on regional functional spillovers beyond the city clusters' boundaries in the producer service sector and the consequent economic outcomes. The findings support the economic prowess of PURs with respect to promoting regional functional spillovers, but only with large populations and great infrastructure connectivity, and the labour division among cities could be the essential mechanism for the linkage between polycentricity and regional functional spillovers.

The remainder of the paper is structured as follows. The following section has a literature review and hypothesis. The third section describes the data and methodology. The empirical results and mechanism discussion are presented in the fourth section. The final section concludes and draws policy implications.

LITERATURE REVIEW AND THEORETICAL HYPOTHESIS

Literature review

Compared with a monocentric urban region, where most of the population is concentrated in one centre, PURs appear to harm agglomeration due to the smaller size of single cities and the longer distances between them (Ciccone, 2002). However, these cities are not isolated from each other. In the literature, PURs are often believed to reap economic advantages through the so-called 'borrowed size' effect among the constituent centres (Meijers & Burger, 2010). This concept was first proposed by Alonso (1973), who noticed that 'a small city or a metropolitan area exhibits some characteristics of a larger one if it is

near other population concentrations'. Alonso believed that small cities close to large cities have higher population potential and can also use public and high-level facilities in large cities such as business services. In other words, small cities could 'borrow' the size or economic advantages of large cities through the interaction between the cities of different scale without increasing costs (de Vos et al., 2020). Later, Parr (2002) expressed the similar view that the agglomeration costs but not the agglomeration benefits are confined more to the boundaries of large cities.

In the following years, the concept of 'borrowed size' has been raised again and extended, but it has often been interpreted with slight differences. Camagni et al. (2017) demonstrated that it was necessary to distinguish between 'borrowed size' and 'borrowed function'. In this regard, 'borrowed size' is expected to bring benefits to large cities because large cities can better use the large market size from their surrounding small cities, while the 'borrowed function' effect mainly brings more advantages to small cities because they can use high-level functions from large cities. The accessibility of the nearby city features (borrowed function) gives all cities an edge, regardless of their size. Meijers and Burger (2017) divided the concept of 'borrowed size' into two dimensions: 'borrowed function' and 'borrowed performance'. A city maintains a higher level of function that is usually not found in cities of its size, and it is defined as a 'borrowed function' because its maintenance is supported by the nearby cities. The nearby support lending cities can also gain access to these functions and other agglomeration benefits to perform better than they would in isolation, which is called 'borrowed performance'.

In contrast to the above-discussed 'borrowed size' effect, which emphasizes the positive effects among cities, there is the 'agglomeration shadows' concept (Meijers et al., 2016), which was originally discussed by the new economic geography (Krugman, 1993). Regarding one kind of metropolitan function, the functions of some cities are greater than expected given their population sizes, and they enjoy the borrowed size effect; however, other cities have fewer functions relative to their sizes and suffer the agglomeration shadow effect within a region (Meijers & Burger, 2017). Empirically, Burger et al. (2015) found that in Northern and Western Europe, larger cities cast a shadow over their smaller neighbouring cities, as opposed to these smaller cities borrowing size from their larger neighbours in regard to high-end cultural amenities.

City network theory is closely related to the conception of 'borrowed size', as both of their key concepts focus on functional interactions among cities (Camagni & Salone, 1993; Capello, 2000; Meijers, 2005; Van Oort et al., 2010). Some factors are essential for the creation of city networks. First, based on means of transportation and information and communication technologies (ICTs), the connections and interactions among independent cities are developed through functional complementarity and synergy (Batten, 1995; Camagni & Salone, 1993). Some scholars have even suggested that network accessibility is more important than geographical proximity

(Johansson & Quigley, 2004; Meijers et al., 2018). Second, the city network is essentially related to interfirm interactions, which make it possible for the spatial division of labour in a PUR (Camagni & Salone, 1993; Van Oort et al., 2010). As the city network paradigm claims, through participation in the network, cities exploit agglomeration economies through complementary relationships and synergies in cooperative activities (Camagni et al., 2017; Capello, 2000). Thus, a PUR can reach a sufficient critical mass for an agglomeration economy in the form of an urban (city) network externality (Capello, 2000), which is considered the essence of city network outcomes or external economies (Burger & Meijers, 2016; Meijers & Burger, 2017).

Although Alonso (1973), Camagni et al. (2017) and Meijers and Burger (2017) interpreted 'borrowed size' differently and although the concept has even been expressed using similar terms such as network externality, all such effects, including the agglomeration shadow effect, are realized through functional interactions between firms and populations among cities (Meijers & Burger, 2017; Phelps et al., 2001). Namely, borrowed size effect and the agglomeration shadow effect describe possible outcomes of interactions of a single city in city networks, and only when the total borrowed size effects dominate the total agglomeration shadow effects can a PUR as a whole achieve positive regional functional spillovers and obtain economic benefits (Meijers & Burger, 2017).

Compared with the rich theoretical literature, empirical evidence on whether PURs have advantages in acquiring positive functional spillovers and economic benefits is scarce (Meijers & Burger, 2010). Most of the related empirical research relies solely on exploring the impact of 'borrowed size' or agglomeration shadows on individual cities (Camagni et al., 2017; Yao & Song, 2019), paying less attention to regional functional spillovers and the economic performance of the PUR as a whole. Focusing only on the impact in individual cities is insufficient to illustrate the significance of network externalities for a region, and the economic rationale and policy relevance of PURs cannot be supported. The measurement of urban functions and performance is also needed at the city network, urban region and local levels (Meijers & Burger, 2017). In this regard, Meijers and Burger (2017) made an advance. They observed that the correlation between the size and function of cities is much lower in a PUR compared with a monocentric urban region and concluded that the borrowed size effect seems to occur more frequently in PURs. However, they still did not directly deal with this issue through rigorous examination. Furthermore, the mechanism behind the city network externalities, which are believed to represent the economic prowess of PURs, has yet to be explored.

Our hypotheses

The insufficient empirical evidence on the city network externalities of PURs motivated us to conduct this empirical study. First, we present our hypotheses. In contrast to a monocentric urban region, a PUR's economic prowess lies

in lower agglomeration diseconomy and more functional spillovers among cities. A PUR has economic advantages over a monocentric urban region only when its economic prowess outweighs the greater agglomeration economy of the monocentric urban region resulting from the larger size of its main centre. Limited by data and methods, this paper cannot afford to examine this grand hypothesis but focuses only on the following two concrete hypotheses aiming at providing empirical evidence on the link from polycentricity through functional spillovers to economic performance, conditional on population size and transportation connectivity. Both our hypotheses and the later empirical work are based on the case of the producer service sector and its division of labour with the manufacturing sector owing to their functional complementarity.

Hypothesis 1: A PUR can promote the deepening of labour division and cooperation among cities by providing more profitable production locations than monocentric urban areas, thus achieving more regional functional spillovers that benefit economic performance.

In monocentric urban regions, the labour division between the producer service sector and the manufacturing sector mainly occurs in large cities instead of among cities because most of the regional production factors are concentrated in large cities. Though small cities generally have a lower land cost and lower worker wages, manufacturing enterprises will not migrate to them due to their insufficient labour force, market and infrastructure and poor transportation connectivity with large cities. In addition, the productive service sector and the manufacturing sector in small cities suffer from the agglomeration shadow effect owing to spatial competition (Burger et al., 2015).

With the rise of the regional polycentricity, the relative population size of small cities and their consequent agglomeration effects will increase. As most types of metropolitan functions are still supported by local size (Meijers et al., 2016), manufacturing enterprises, which are usually cost sensitive, will begin to relocate to these subcentre cities, while the producer services sector will tend to continue to cluster in large cities. The labour division between the producer services sector and the manufacturing sector that originally occurred within large cities will shift to among cities.

The manufacturing sector located in growing small cities can perform better owing to the effect of increasing returns to scale and a relatively lower agglomeration diseconomy, and more demand for producer services will be generated. The regional agglomeration of the producer services sector increases the supply capacity. Thus, the labour division and functional interactions between the manufacturing sector in small cities and producer services in large cities will be enhanced in general, and the so-called 'borrowed size' effect will be increased. Moreover, the division of labour among cities also helps to decrease the agglomeration shadow effect on the producer services or manufacturing function because they are located in different cities.

Hypothesis 2: The promoting effects of polycentricity on the spatial division of labour and the consequent functional spillovers are conditional on the interaction opportunities among cities inside a PUR. A larger population size and greater infrastructure connectivity are essential for facilitating interactions and supporting the division of labour among cities inside a PUR.

Market scale (population) and transportation facilities are believed to have an important impact on the division of labour (Smith, 2010). Coincidentally, Camagni and Salone (1993) also claimed that the original definition of urban networks includes infrastructure systems and spatial interactions. A larger population or market scale means more interactions and trading opportunities, which are conducive to the deepening of the division of labour and that among cities inside a PUR since different sectors have different location preferences to pursue the maximization of income. In addition to a large population size, a well-connected transportation infrastructure is essential to support the labour division and functional interactions among cities. Due to greater transportation connectivity, the producer services sector and manufacturing sector in different cities inside a PUR transact with the lowest costs just as if they were in the same city, and all cities inside the PUR could make intense use of the network, which would strengthen the network externality or agglomeration economy in general (Capello, 2000).

To summarize, given that different cities (locations) are suitable for different economic functions (Van Oort et al., 2010), polycentricity promotes the deepening of the division of labour and the formation of a city network among cities. The division of labour captures the essence of the 'borrowed size' effect and alleviates the 'agglomeration shadow effect'. Consequently, regional function spillovers increase, which benefits regional economic performance. The spatial strategy of enterprises plays an essential role in the formation of city networks (Camagni & Salone, 1993; Van Oort et al., 2010).

The shift in the division of labour from inside large cities to among cities is in accordance with the spatial restructuring process of regional industrialization (Friedmann, 1966), during which most economic sectors concentrate in the major centre in the first phase, and spatial division occurs in the later phase and is accompanied by morphological polycentricity. Therefore, the above hypotheses are more applicable to the context of regional industrialization such as occurs in China (Chen et al., 2006). Whether these hypotheses are generalizable to post-industrialized economies is worth verifying in the future (Phelps & Ozawa, 2003).

DATA AND METHOD

In order to test our hypotheses, the estimation strategy of this paper is to examine the positive effect of polycentricity on functional spillovers and the consequent economic performance at first, and then explore the mechanism linking the polycentricity and functional spillovers, namely the labour division.

Research areas and data

The research sample consists of 13 Chinese city clusters, which are relatively well known in China and which were identified by Ning (2016). Regarding the data source, the population data of each city are gathered from Landscan (Dobson et al., 2000; Liu et al., 2018). The frequency of trains for each city within city clusters is collected from the China train schedule published by the China Railway Corporation. Other data, including the number of internet users, the number of employees, gross domestic product (GDP), total investment in fixed assets, the number of college students and government consumption, are collected from the China city statistical yearbook. All data are from the period 2004–14, with a gap every two years.

The measurement of the mono/polycentricity of city clusters

Following most of the previous literature (Meijers, 2008; Meijers & Burger, 2010; Sun et al., 2017), we measure the morphological mono/polycentricity of city clusters based on their city population rank-size distribution as follows:

$$\ln(\text{Rank} - 0.5) = a + \partial \ln \text{Size} + \varepsilon \quad (1)$$

where Rank is the rank of the cities according to their population in a city cluster. We subtract 0.5 from Rank to reduce the potential sample size bias for the Pareto exponent (Gabaix & Ibragimov, 2011). Size means the city population size. The coefficient ∂ is the Pareto exponent, representing the level of morphological polycentricity. To explain ∂ in equation (1) more easily, ∂ or the Pareto index mentioned below is the absolute value of ∂ . The higher the Pareto index is, the more polycentric the city cluster. a and ε are the constant and error terms, respectively (Table 1).

The measurement of the functional spillovers of a city cluster in the producer service sector

We choose the producer service sector as the research object of functional spillovers for two reasons. First, the producer service sector is playing an increasingly important role in the modern economy; thus, its performance is a good indicator of regional economic development. Second, with the deepening of the division of labour among cities, the producer service sector, which is functionally specialized, often agglomerates in some cities and serves the manufacturing sector in other cities because they have the production linkage and a complementarity network is thus formed. The producer service sector in this paper includes transportation, warehousing and postal services, information transmission, computer services and the software industry, financial services, the real estate industry, leasing and business services, and scientific research, technical services and geological exploration (Chen et al., 2009).

The procedure for measuring the functional spillovers of a PUR beyond its boundary consists of four steps. First, the employment share of the producer service sector of each city compared with the total

Table 1. Characteristics of the 13 city clusters in 2014.

City cluster	Number of cities included	Population size (×1000)	Polycentricity	City cluster	Number of cities included	Population size (×1000)	Polycentricity
Central Plains	5	29,576.292	2.422	Pearl River Delta	9	56,841.152	1.216
Changchun-Jilin	2	12,355.282	1.971	Yangtze River Delta	15	102,938.639	1.190
Chang-zhu-tan	3	13,864.109	1.717	Jing-jin-ji	5	48,242.533	1.074
Mindongnan	5	26,039.169	1.662	Chengdu-Chongqing	5	53,875.884	0.815
Shandong Peninsula	6	39,217.211	1.556	Guangzhong	4	18,412.486	0.722
Liaozhongnan	7	26,871.16	1.393	Wuhan	3	13,495.167	0.721
Ha-da-qi	3	19,211.908	1.245				

employment of the producer service sector in the city cluster is defined as an indicator of the city's actual level of the producer service sector. Second, referring to Meijers and Burger (2017), Meijers et al. (2018) and Volkmann and Rusche (2019), we regress the proportion of the population of the city in the city cluster on its producer service sector to obtain the expected level of the employment share of the city equal to its population size share. Other variables that might determine the level of the producer service sector are also controlled, including the city's manufacturing employment share compared with that of the city cluster, human capital (the number of college students per 10,000 persons), innovation (the ratio of research input compared with that of the city cluster), government intervention (the ratio of government consumption to urban *GDP*, subtracting education and research input), and informatization (the number of mobile phones per capita). Third, we calculate each city's regression residual, which is considered the discrepancy between its actual level of the producer service sector and its expected level. The higher the absolute value of the residual is, the greater the mismatch in the relation between the producer service function and its corresponding population size. We use the residuals as an indicator of the functional performance of a city in the producer service sector, referring to Meijers and Burger (2017), Meijers (2008) and Volkmann and Rusche (2019). More specifically, if the residual of a city is positive, the city is a producer service sector centre (city) in the city cluster and enjoys the 'borrowed size' effect because its producer service function goes beyond the demand equal to its size and tends to borrow demand size for the producer service function from other cities within the city cluster (Meijers & Burger, 2017). In contrast, if the residual is negative, the city suffers the agglomeration shadow effect in the producer service sector (Meijers & Burger, 2017) and tends to borrow functions from producer service sector centres. Fourth, we add the residuals of the cities in each city cluster as the functional spillovers of the city cluster in the producer service sector, reflecting the trade-offs between borrowed size and agglomeration shadows among cities. If the sum is positive, the borrowed size exceeds the agglomeration shadow, and the

city cluster's productive service function can not only meet the needs of its own city cluster compared with its size but also spill over to the regions beyond the city cluster's boundary.

Estimation model for the effect of polycentricity on functional spillovers and economic performance

The basic logic of our study is that polycentricity promotes the regional functional spillovers of PURs, which further benefits economic performance. The most direct method is to introduce two separate equations to test the above two relationships independently (as in equations 2 and 3). However, functional spillovers ($\ln FS_{i,t}$) are obviously endogenous, as functional spillovers are the dependent variable not only in equation (2) but also on the right-hand side of equation (3). Therefore, we adopt the simultaneous equation model to estimate all the equations as a whole to examine the effects of polycentricity on functional spillovers (equation 2) as well as the effects of functional spillovers on economic performance (equation 3). The over-identification of both equations (2) and (3) supports the feasibility of the simultaneous equation model. Additionally, we adopt three-stage least squares (3SLS, a special kind of simultaneous equation method using full information) to estimate the model. Compared with ordinary least squares (OLS) and two-stage least squares (2SLS), the 3SLS method further takes into accounts the disturbance terms of each equation and is more efficient in parameter estimation than OLS and 2SLS. Our discussions below are mainly based on 3SLS results, although we retain OLS and 2SLS results for comparison.

$$\begin{aligned}
 \ln FS_{i,t} = & \beta_1 + \beta_2 \ln PARETO_{i,t} + \beta_3 \ln POP_{i,t} \\
 & + \beta_4 \ln INF.CONNECT_{i,t} + \beta_5 ADMIN_{i,t} \\
 & + \beta_6 \ln INTERNET_{i,t} + \beta_7 \ln DIVERSITY_{i,t} \\
 & + \beta_8 CENTRAL_{i,t} + \beta_9 WESTERN_{i,t} \\
 & + \beta_{10} \ln PARETO_{i,t} * \ln POP_{i,t} \\
 & + \beta_{11} \ln PARETO_{i,t} * \ln INF.CONNECT_{i,t} + \varepsilon_{i,t}
 \end{aligned} \quad (2)$$

$$\begin{aligned} \ln GDP_{i,t} = & \beta_1 + \beta_2 \ln FS_{i,t} + \beta_3 \ln HUMAN_{i,t} \\ & + \beta_4 \ln INVEST_{i,t} + \beta_5 \ln POP_{i,t} \\ & + \beta_6 CENTRAL_{i,t} + \beta_7 WESTERN_{i,t} + \varepsilon_{i,t} \end{aligned} \quad (3)$$

In equation (2), i represents the city cluster; t represents the year; FS means functional spillovers; and $PARETO$ (henceforth, the absolute value of ∂) is the indicator of the spatial polycentricity of city clusters. The higher is $PARETO$, the more polycentric the city cluster. We expect the coefficient β_2 of $PARETO$ to be positive. POP is the population of the city cluster. Potentially, the number of cities included in a city cluster affects the measurement of the spatial structure (Zhang & Derudder, 2019). Given that the number of cities and the population size are collinear, we consider only the population in the model. $INF_CONNECT$ is an indicator of the infrastructure connectivity within the city cluster, measured by the average frequency of trains in each city within the city cluster. Similarly, Meijers et al. (2018) also used frequency train connections (average number of trains between each pair of cities in a PUR) to measure the regional functional coherence. The higher train frequency could increase the utilization of infrastructure and connectivity with other cities and reduce the inter-city transaction cost. The measure of $INF_CONNECT$ is as follows, and we assume that there are n cities in the city cluster, and i, j mean different city. TF_{ij} is the train frequency (TF) between city i and city j :

$$INF_CONNECT = \frac{\sum_{i \neq j}^n TF_{ij}}{2n} \quad (4)$$

$ADMIN$ refers to the number of provinces included in the city cluster; it is used to control for the potential provincial boundary frictions affecting functional spillovers. Sohn and Licheron (2017) confirmed the ambivalent effects of borders on metropolitan functions in Europe. $INTERNET$ refers to the number of internet users per capita in the city cluster and reflects the effect of the internet on communication among cities. $DIVERSITY$ refers to the industrial diversity of the city cluster except for the productive service sector, and it is calculated by equation (5). Industrial diversity is not conducive to functional spillovers to the producer service sector to a certain extent. In equation (5); $Y_{i,k}$ is employment in sector k in city cluster i ; and $Y_i - Y_{i,p}$ is employment in all sectors (Y_i) except the productive service sector ($Y_{i,p}$), in city cluster i :

$$DIVERSITY = \frac{1}{\sum_{k \neq p} (Y_{i,k} / (Y_i - Y_{i,p}))^2} \quad (5)$$

$CENTRAL$ and $WESTERN$ are dummy variables that capture broad regional fix effects. We also introduce some interaction terms to examine whether the effects of polycentricity on functional spillovers vary with some of the city cluster attributes.

The Cobb–Douglas production function is commonly used in estimating labour productivity, which is applied for equation (3). Labour productivity is measured by GDP per worker. The functional spillovers (FS) is expected to be positively correlated with labour productivity. For the factor inputs, investment and human capital are expected to have positive coefficients for labour productivity. $HUMAN$ means human capital, measured by the ratio of college students per 10,000 workers, to be consistent with the denominator (worker) of labour productivity. $INVEST$ denotes the total investment in fixed assets per worker within the city cluster. Population size (POP) is controlled to reflect the effect of the agglomeration economy. $CENTRAL$ and $WESTERN$ are regional dummy variables (Table 2).

To test whether it is appropriate to use the simultaneous equation model, we regress the residual e of equations (2) and (3). The coefficient of e is not zero, and its significant level is 10% ($p = 0.058$), which means that the residual of equation (2) has a significant effect on equation (3) and thus supports the necessity of applying the simultaneous equation model.

Based on the source of standard deviations (Table 3), the variances of both core variables, polycentricity ($PARETO$) and functional spillovers (FS), are mainly from the ‘between’ variation of different city clusters, rather than the ‘within’ variation of the same city cluster in time. Thus, we cease controlling the city cluster fixed effect (Wooldridge, 2016).

Estimation model for the effect of polycentricity on functional spillovers through labour division among cities

This part explores why polycentricity facilitates functional spillovers, and we believe that the division of labour among cities plays an essential role.

The first step is measuring the division of labour among cities. Greater division of labour is positively related to the dissimilarity of the industrial structures of cities, which can be effectively measured by grey relational analysis (GRA) (Li & Miao, 2006). The higher a city’s grey relational grade (GRG) is, the more similar the industrial structure of the city is to that of the reference city. Thus, the coefficient of variation (CV) of the GRG among cities is used as an indicator to measure the division of labour among cities. It is supposed that a higher CV means that the industrial structures of cities are less similar. The processes used for the application of GRA and the calculation of GRG here follow the processes proposed by Li and Miao (2006).

The second step is to use the simultaneous equation model and 3SLS again, and we still cease controlling for the city cluster fixed effect for the same reason that the standard deviation of $\ln CV$ is mainly from the ‘between’ (0.053) variation of different city clusters rather than the ‘within’ (0.042) variation of the same city cluster in time (Wooldridge, 2016). The over-identification of both equations (6) and (7) supports the feasibility of the simultaneous equation model. The residual of equation (6) has a significant effect on equation (7) with a significance level

Table 2. Descriptive statistics of the variables.

	Definition	Mean	SD	Minimum	Maximum
<i>Endogenous variables</i>					
lnFS	Sum of residuals of the constituent cities in each city cluster	0.074	0.136	−0.125	0.324
lnGDP	Gross domestic product (GDP) per worker (10,000 yuan per worker)	2.453	0.376	1.510	3.320
<i>Exogenous variables</i>					
lnPOP	Number of inhabitants in a city cluster (10,000)	7.918	0.605	7.103	9.240
lnPARETO	Indicator for the spatial polycentricity of city clusters (the absolute value of ϑ)	0.850	0.209	0.494	1.231
lnINF_CONNECT	Indicator of infrastructure connectivity measured by the average train frequency of each city within the city cluster	3.915	1.048	0.336	6.237
lnINTERNET	Number of internet users per capital in the city cluster	0.150	0.084	0.038	0.523
ADMIN	Number of included provinces in the city cluster	1.385	0.743	1	3
WESTERN	Dummy variable: the western China city cluster	0.154	0.363	0	1
CENTRAL	Dummy variable: the central China city cluster	0.231	0.424	0	1
lnDIVERSITY	Industrial diversity of the city cluster	1.524	0.436	0.190	3.921
lnHUMAN	Ratio of college student per 10,000 workers	6.567	0.516	5.458	7.416
lnINVEST	Total investment in fixed assets per worker within this city cluster (10,000 yuan per worker)	2.144	0.571	0.762	3.066

of 5% ($p = 0.026$).

$$\begin{aligned} \ln CV_{i,t} = & \beta_1 + \beta_2 \ln PARETO_{i,t} + \beta_3 \ln POP_{i,t} \\ & + \beta_4 \ln INF_CONNECT_{i,t} + \beta_5 \ln GOV_{i,t} \\ & + \beta_6 CENTRAL_{i,t} + \beta_7 WESTERN_{i,t} \\ & + \beta_8 \ln PARETO_{i,t} * \ln POP_{i,t} \\ & + \beta_9 \ln PARETO_{i,t} * \ln INF_CONNECT_{i,t} + \varepsilon_{i,t} \end{aligned} \quad (6)$$

$$\begin{aligned} \ln FS_{i,t} = & \beta_1 + \beta_2 \ln CV_{i,t} + \beta_3 \ln POP_{i,t} \\ & + \beta_4 \ln INF_CONNECT_{i,t} + \beta_5 ADMIN_{i,t} \\ & + \beta_6 \ln INTERNET_{i,t} + \beta_7 \ln DIVERSITY_{i,t} \\ & + \beta_8 CENTRAL_{i,t} + \beta_9 WESTERN_{i,t} \\ & + \beta_{10} \ln CV_{i,t} * \ln POP_{i,t} \\ & + \beta_{11} \ln CV_{i,t} * \ln INF_CONNECT_{i,t} + \varepsilon_{i,t} \end{aligned} \quad (7)$$

Table 3. Source of variation of polycentricity (*PARETO*) and functional spillovers (*FS*).

	Mean	SD	Minimum	Maximum
PARETO(ln)				
Overall	0.850	0.209	0.494	1.231
Between		0.211	0.542	0.939
Within		0.041	0.609	0.939
FS(ln)				
Overall	0.074	0.136	−0.125	0.324
Between		0.126	−0.055	0.277
Within		0.060	−0.101	0.267

where i is the city cluster; and t is the year. In equation (6), the explained variable is $CV_{i,t}$ (the coefficient of the variation of the GRG among cities), and the core explanatory variable is polycentricity ($PARETO_{i,t}$). We consider that the two variables are positively correlated. Regional government intervention $GOV_{i,t}$ (the ratio of city cluster government consumption to city cluster GDP) is controlled for. The other control variables are shown in equation (6), which is provided in Table 5. In equation (7), the interpreted variable is regional function spillovers $FS_{i,t}$, and the core explanatory variable is $CV_{i,t}$, which is expected to be positively correlated with $FS_{i,t}$. The other control variables are consistent with equation (2). In this part, the new variables are the coefficient of variation ($CV_{i,t}$) and regional government intervention ($GOV_{i,t}$), so the descriptive statistics of the other variables are not reported here. $\ln GOV$'s mean is 0.084, and its minimum and maximum are 0.042 and 0.152, respectively. The standard deviation of $\ln GOV$ is 0.024. $\ln CV$'s mean is 0.226, and its minimum and maximum are 0.067 and 0.390, respectively. The standard deviation of $\ln CV$ is 0.067.

RESULTS

Spatial structure, regional functional spillovers and regional labour productivity

We conduct more elaborate regressions to obtain robust results (Table 4). In each model, the upper part focuses on the effect of polycentricity on functional spillovers, corresponding to equation (2); the lower part deals with the influence of functional spillovers on labour productivity, corresponding to equation (3).

Models (1)–(3) present the OLS, 2SLS and 3SLS estimation results, respectively, with polycentricity having a constant effect on functional spillovers. Considering the possibility that the spillover process could depend on the

population potential and transportation cost or infrastructure connectivity of city clusters, we introduce the interaction terms between polycentricity and corresponding attributes of city clusters in models (4)–(6). To make the

Table 4. Spatial structure, regional functional spillovers and regional labour productivity.

	(1) OLS	(2) 2SLS	(3) 3SLS	(4) OLS	(5) 2SLS	(6) 3SLS
FS(ln)						
<i>PARETO</i> (ln)	0.082 (1.23)	0.082 (1.23)	0.017 (0.29)	0.311*** (3.51)	0.311*** (3.51)	0.265*** (3.38)
<i>POP</i> (ln)	0.183*** (6.18)	0.183*** (6.18)	0.192*** (7.31)	0.152*** (5.04)	0.152*** (5.04)	0.159*** (5.97)
<i>INF_CONNECT</i> (ln)	−0.001 (−0.09)	−0.001 (−0.09)	−0.006 (−0.50)	−0.016 (−1.16)	−0.016 (−1.16)	−0.016 (−1.31)
<i>ADMIN</i>	−0.151*** (−7.20)	−0.151*** (−7.20)	−0.155*** (−8.29)	−0.103*** (−4.49)	−0.103*** (−4.49)	−0.112*** (−5.53)
<i>INTERNET</i> (ln)	0.104 (0.54)	0.104 (0.54)	0.036 (0.22)	0.117 (0.66)	0.117 (0.66)	0.094 (0.60)
<i>DIVERSITY</i> (ln)	−0.031 (−1.12)	−0.031 (−1.12)	−0.033 (−1.41)	−0.021 (−0.82)	−0.021 (−0.82)	−0.023 (−1.04)
<i>CENTRAL</i>	−0.034 (−1.18)	−0.034 (−1.18)	−0.032 (−1.21)	−0.104*** (−3.14)	−0.104*** (−3.14)	−0.097*** (−3.31)
<i>WESTERN</i>	−0.092** (−2.27)	−0.092** (−2.27)	−0.117*** (−3.30)	−0.064* (−1.66)	−0.064* (−1.66)	−0.076** (−2.22)
<i>POP</i> (ln) _c * <i>PARETO</i> (ln) _c				0.327** (2.20)	0.327** (2.20)	0.281** (2.15)
<i>INF_CONNECT</i> (ln) _c * <i>PARETO</i> (ln) _c				0.132* (1.90)	0.132* (1.90)	0.127** (2.08)
<i>YEAR</i>	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Constant	−1.175*** (−5.48)	−1.175*** (−5.48)	−1.155*** (−6.06)	−1.145*** (−5.26)	−1.145*** (−5.26)	−1.144*** (−5.92)
<i>R</i> ²	0.620	0.620	0.614	0.693	0.693	0.691
GDP per worker(ln)						
<i>FS</i> (ln)	0.318*** (3.09)	0.570*** (3.90)	0.571*** (4.25)	0.318*** (3.09)	0.405*** (3.16)	0.408*** (3.46)
<i>HUMAN</i> (ln)	0.336*** (6.81)	0.355*** (6.81)	0.324*** (7.28)	0.336*** (6.81)	0.343*** (6.86)	0.324*** (7.13)
<i>INVEST</i> (ln)	0.122*** (2.64)	0.102** (2.08)	0.141*** (3.33)	0.122*** (2.64)	0.115** (2.45)	0.130*** (3.05)
<i>POP</i> (ln)	0.105*** (3.15)	0.098*** (2.81)	0.087*** (2.81)	0.105*** (3.15)	0.103*** (3.06)	0.095*** (3.08)
<i>CENTRAL</i>	−0.350*** (−9.99)	−0.341*** (−9.28)	−0.336*** (−10.00)	−0.350*** (−9.99)	−0.347*** (−9.82)	−0.343*** (−10.57)
<i>WESTERN</i>	−0.552*** (−14.29)	−0.523*** (−12.46)	−0.516*** (−13.45)	−0.552*** (−14.29)	−0.542*** (−13.61)	−0.537*** (−14.69)
<i>YEAR</i>	Controlled	Controlled	Controlled	Controlled	Controlled	Controlled
Constant	−1.142** (−2.17)	−1.209** (−2.19)	−0.968** (−2.04)	−1.142** (−2.17)	−1.165** (−2.20)	−1.002** (−2.07)
Observations	78	78	78	78	78	78
<i>R</i> ²	0.929	0.923	0.925	0.929	0.928	0.928

Note: *t*-statistics are given in parentheses. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

effects more interpretable, we subtract the mean from each variable and then compute the interaction terms.

Table 4 shows that the findings in our most reliable 3SLS model (models 3 and 6) are consistent with those of 2SLS model (models 2 and 5), compared with the weak significance of functional spillovers in the GDP per worker model by simply using the OLS model (lower part of models 1 and 4).

The results in model (3) indicate that polycentricity does not have a direct relationship with functional spillovers when taking no account of the moderating effect of other factors. However, as model (6) shows, the polycentric spatial structure could boost the functional spillovers of the producer service sector for the city cluster, which is 'average' in terms of the population size and infrastructure connectivity (as we centre the variables before we compute the interaction terms). Moreover, the positive signs of two interaction terms ($POP(\ln)*PARETO(\ln)$ and $INF_CONNECT(\ln)*PARETO(\ln)$) suggest that both population size and infrastructure connectivity could enhance the effect of polycentricity on functional spillovers. Due to an increase in the total population size of city clusters, the opportunities for interactions grow concomitantly, which boosts the borrowed size effect of polycentricity. Increased infrastructure connectivity could reduce the transaction costs caused by the distance friction and space barriers between cities to some extent and enhance the convenience of flow. The higher the frequency of trains is, the better the utilization of infrastructure and the greater the possibility that the working population will interact with others within the city cluster.

Furthermore, we calculate each marginal effect of polycentricity conditional on different levels of population and infrastructure connectivity of the city clusters. As Figure 1 shows, polycentricity is more amenable to functional spillovers only in large (more than 18.08 million people) and closely connected (a frequency of more than 24 trains per day) city clusters. Regarding economic performance, the

results in the lower part of both models (3) and (6) show that the positive effects of functional spillovers are quite robust.

With respect to other control variables, we find a positive effect of population on functional spillovers, which means that city clusters with large populations have more potential to spill over. This is in line with Alonso's (1973) population potential theory. We also find that there are segmentation effects of provincial boundaries in China's city clusters because provincial boundaries (*ADMIN*) have a negative impact on functional spillovers. We conclude that administrative barriers interfere with the functional spillovers of producer service sector. Moreover, city clusters in the central and western regions of China have fewer functional spillovers than do those in the eastern region.

Higher labour productivity is accompanied by more functional spillovers (*FS*) in all models as expected. Because the producer service function is playing an increasingly important role in the modern economy, thus its better performance is relative to better regional economic development. Investment and human capital are positively associated with the economic performance, which are consistent with the economic theories. Population and economic performance are positively correlated, which is in line with agglomeration economic theory.

Further analysis: spatial structure, the division of labour and regional functional spillovers

Based on the hypothesis mentioned in the second section, Labour division among cities plays a critical role in the linkage between polycentricity and regional functional spillovers. To test the above mechanism, we conducted a quantitative analysis. The regression results are shown in Table 5. The upper part focuses on the effect of polycentricity on the division of labour, corresponding to equation (6); the lower part addresses the influence of the division of labour on functional spillovers, corresponding to equation (7).

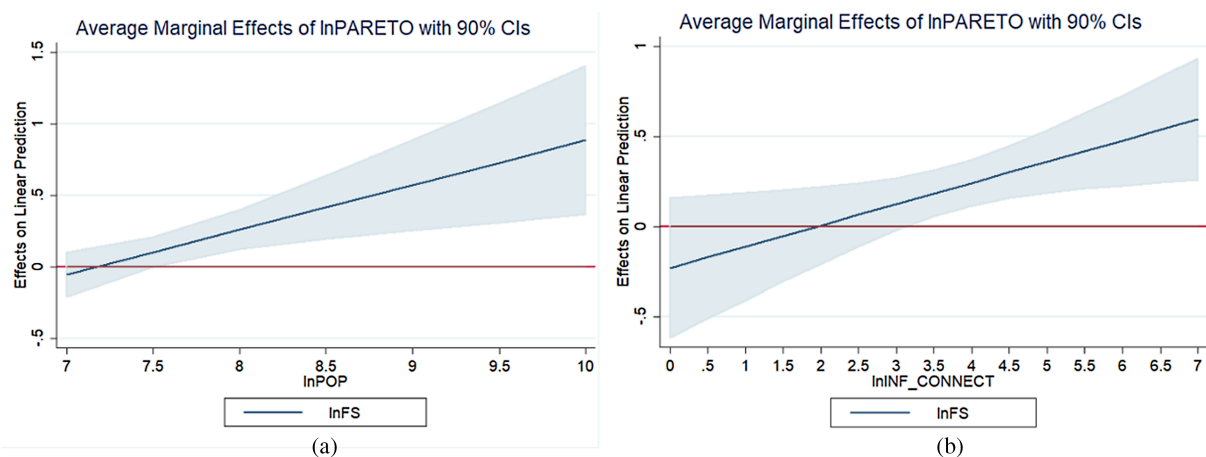


Figure 1. Marginal effects of $\ln PARETO$ (based on model 4 in Table 4).

Note: Moderating effects of population (a); and moderating effects of infrastructure connection (b).

By comparing models (1) and (2) in Table 5, it can be seen that when the market size or population grows, the development of polycentricity will promote industrial dissimilarity between cities, that is, deepen the division of labour among cities. The division of labour among cities can enhance functional spillovers under the condition that the infrastructure is better connected. Therefore, a polycentric spatial structure could conditionally boost the functional spillovers of the producer service sector. The result in Table 5 echoes the results in Table 4. This result further indicates that the division of labour and cooperation among cities could be part of the essential driving forces of ‘borrowed size’ effect.

Table 5. Spatial structure, division of labour and regional functional spillovers.

	(1) 3SLS	(2) 3SLS
CV(ln)		
POP(ln)	−0.047*** (−2.94)	−0.032** (−2.11)
PARETO(ln)	0.041 (1.46)	0.198*** (4.19)
GOV(ln)	0.466 (1.24)	1.177*** (3.05)
INF_CONNECT(ln)	−0.005 (−0.57)	0.030 (0.92)
POP(ln)_c*PARETO(ln)_c		0.369*** (4.01)
INF_CONNECT(ln)_c*PARETO (ln)_c		−0.052 (−1.38)
REGION	Controlled	Controlled
YEAR	Controlled	Controlled
Constant	0.546*** (4.90)	0.114 (0.56)
R ²	0.2119	0.3557
FS(ln)		
CV(ln)	1.624 (0.57)	0.817** (2.24)
CV(ln)_c*POP(ln)_c		−0.006 (−0.02)
CV(ln)_c*INF_CONNECT(ln)_c		0.614*** (3.40)
VARIABLE	Controlled	Controlled
REGION	Controlled	Controlled
YEAR	Controlled	Controlled
Constant	−2.430 (−1.59)	−1.722*** (−5.83)
Observations	78	78
R ²	0.0899	0.5515

Notes: *t*-statistics are given in parentheses.

p* < 0.1, ** *p* < 0.05, * *p* < 0.01.

CONCLUSIONS

By taking 13 city clusters in China as a sample of PURs and using 3SLS to estimate the simultaneous relationship, we examine the role of the polycentricity of PURs in boosting regional functional spillovers (the trade-off of the borrowed size effect and agglomeration shadows effect) beyond their boundaries and consequent economic performance. We find that polycentricity is more beneficial for regional functional spillovers when city clusters have larger populations and greater urban infrastructure connectivity, which consequently increase the labour productivity of city clusters. That is, the economic prowess of PURs with respect to the often hypothesized ‘borrowed size’ effect is conditional on the population size and infrastructure connectivity of PURs, at least in the context of Chinese city clusters. The division of labour between the producer service sector and the manufacturing sector plays a positive role in the polycentricity’s promoting impact on regional functional spillovers. Our findings directly support our hypotheses and reveal that the division of labour among cities in PURs could be the essential driving force of ‘borrowed size’ in previous literature, at least in the case of producer service functions.

The policy implications of this research are obvious. First, the polycentric spatial planning concept is more appropriate for PURs with a large population size and great infrastructure connectivity, at least in the sense of the ‘borrowed size’ effect. Second, to increase the economic rationale of PURs, it is essential to improve the network connectivity of transport infrastructure among the cities within a PUR.

The findings of this paper call for further research in several issues. First, what needs to be acknowledged is that the mechanism of the promoting effect of polycentricity on functional spillovers is based on the evidence from an emerging country China, where small cities in a monocentric region often have less attractiveness to manufacturing enterprises owing to limited labour pool and market scale and poor infrastructure both insides and outsides with large cities. Whether the conclusion and the mechanism are generalizable to developed or post industrialized economies is worth verifying.

Second, the residual method of measuring ‘borrowed size’ is far from perfect. Due to the omitted variable problem, the residual can approximate the actual degree of ‘borrowed size’, but it is not the actual degree of ‘borrowed size’ itself. There is an error in the residual, although we have controlled the variables as much as possible. If they are available, flow data are more appropriate for measuring ‘borrowed size’, since flow is what really happens in the ‘borrowed size’ process.

Third, in addition to the mechanism of the division of labour between the producer service sector and manufacturing sector analysed in this paper, there may be other possible mechanisms that enhance the promoting effect of polycentricity on the functional spillovers or overall performance of producer services at a regional scale.

DISCLOSURE STATEMENT

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